

PDF CAPABILITY DOSSIER

AAA Agentic Intelligence Skills for Human Artifacts

A single frame for the sovereign: which PDF capabilities exist, who owns them, what gates them, and what actually executes when a human artifact is requested. Every number in this document was probed live at 08:0x MYT 02 Oct 2026 — no assumption, no narration.

Purpose	F13 sovereign reference frame
Scope	PDF skills · AAA agents · arifOS kernel · A-FORGE actuators
Probe method	live binaries · MCP tools/list · registry JSON · filesystem
Sources	8 kernel tools · 15 agents · 32 A-FORGE tools

Executive summary — what this is

- **13 PDF-related skill artifacts confirmed on disk** (probed by path in this session: 6 in the primary [document-intel/](#) workshop tree + 7 scattered leaf directories across Hermes and AAA trees). Every one verified LIVE — SKILL.md present with real content.
- **The 3-layer architecture is real and live.** AAA agents own the workflow, arifOS kernel owns the gates (8 tools: arif_init→arif_seal), A-FORGE owns the actuators (32 tools, 95 OBSERVE / 23 MUTATE).
- **8 PDF engines verified working today** on this VPS: weasyprint, reportlab, pandoc, pymupdf, matplotlib (chart), graphviz dot, chrome headless, libreoffice.
- **Real gaps exist:** no *marker-pdf* (OCR scanned) · no voice-to-PDF single-call · no live-data-into-PDF-while-building. Name them, don't fake them.
- **The deliverable discipline is the handshake.** What renders on PDF reaches the human. What stays in an agent log does not. PDF = the visible handshake.

1 · PDF production pipeline — 7 stages, no skipping

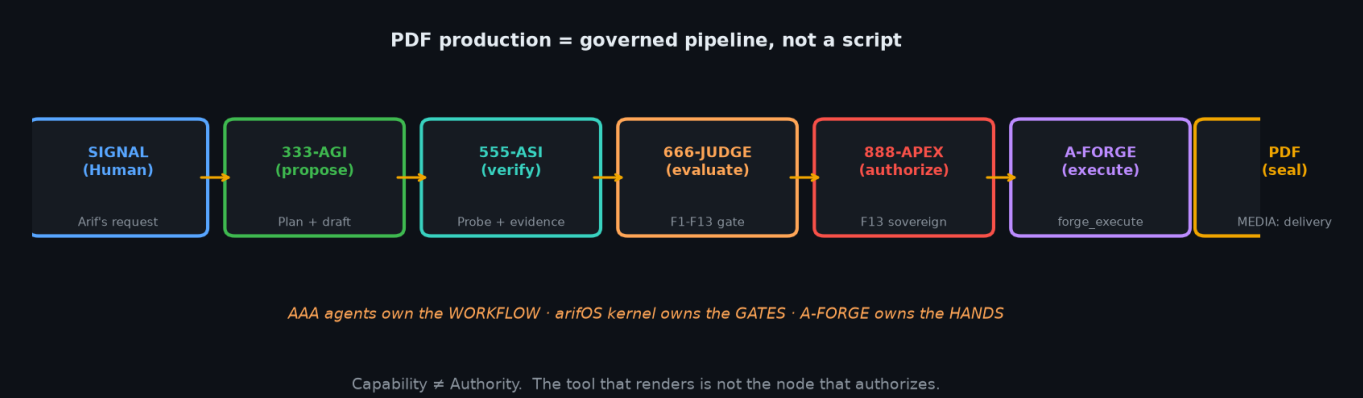


Figure 1 — A PDF is not a script output; it is a governed artifact. Signal enters, the agent proposes, verification gates, F13 authorizes, A-FORGE executes, the artifact reaches the human.

2 · Three-layer architecture — separate powers, one flow

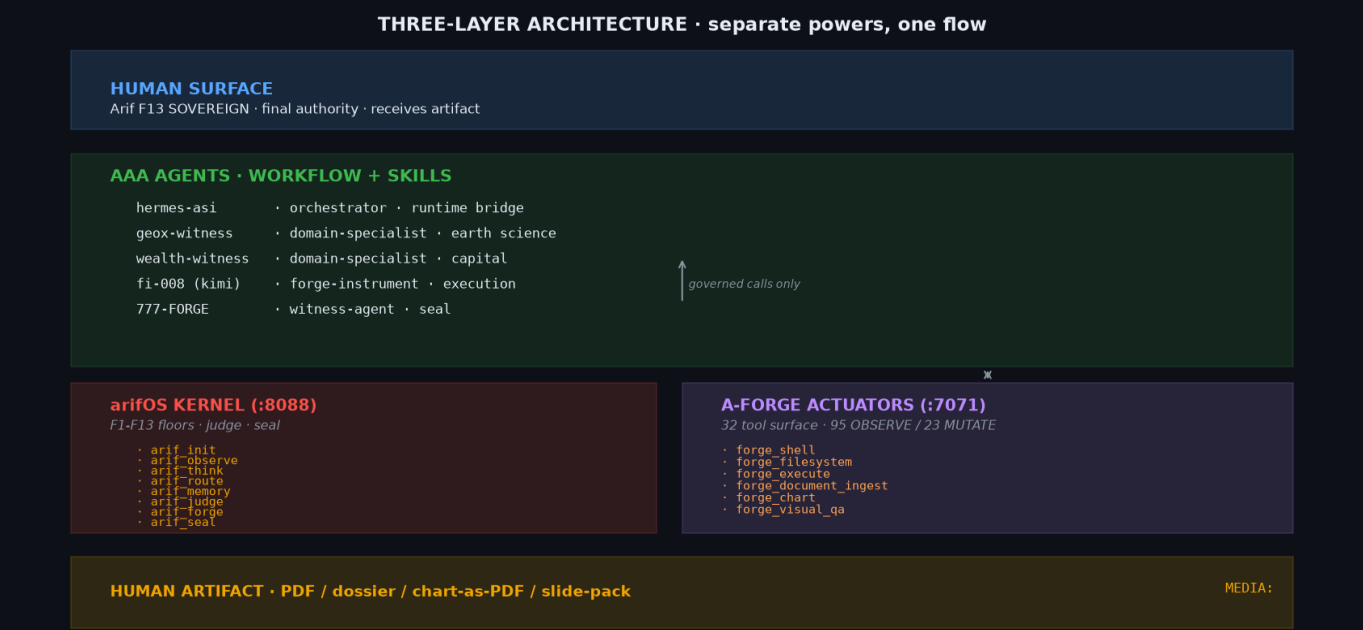


Figure 2 — Capability ≠ Authority enforced structurally. Agents hold skills; the kernel holds authority; A-FORGE holds the hands. No agent issues its own envelope.

3 · PDF skills inventory — verified on disk

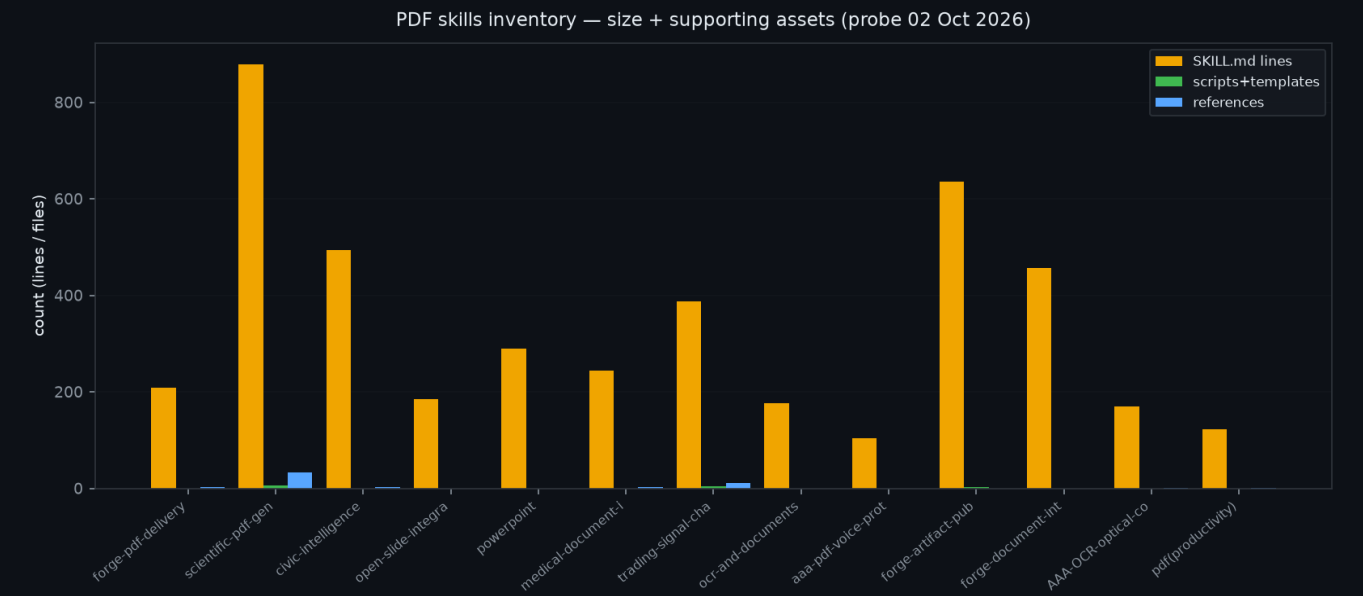


Figure 3 — Bar size = SKILL.md line count; green = scripts+templates; blue = reference files. A stub has lines but no scripts: the skill narrates, it does not execute.

Inventory table — status from filesystem probe

Skill	Status	SKILL.md	scripts+tpl	refs
forge-pdf-delivery	LIVE	209	0	3
scientific-pdf-generation	LIVE	879	6	33
civic-intelligence-pdf	LIVE	493	0	2
open-slide-integration	LIVE	184	0	0
powerpoint	LIVE	289	0	0
medical-document-interp	LIVE	244	0	2
trading-signal-chart	LIVE	387	4	11
ocr-and-documents	LIVE	177	0	0
aaa-pdf-voice-protocol	LIVE	103	0	0
forge-artifact-publisher	LIVE	635	2	0
forge-document-intel	LIVE	457	0	0
AAA-OCR-optical-compress	LIVE	169	0	1
pdf(productivity)	LIVE	122	0	1

4 · Routing matrix — which skill for which job

Routing matrix · green ✓ = use it, ~ = partial, X = don't								
forge-pdf-delivery	✓	✓	X	X	X	X	X	X
scientific-pdf-gen	✓	✓	✓	~	~	X	X	X
civic-intelligence-pdf	~	✓	X	X	~	X	X	X
trading-signal-chart	X	X	X	✓	X	X	X	X
open-slide-integration	~	~	X	X	✓	X	X	X
powerpoint	X	X	X	X	✓	X	X	X
ocr-and-documents	X	X	X	X	X	X	✓	✓
aaa-pdf-voice-protocol	X	~	X	X	X	✓	X	X
forge-artifact-pub	~	✓	~	X	✓	~	X	~
forge-document-intel	X	~	X	X	X	~	~	✓
	Plain text doc	Dark dossier + chart	Sell-side analyst	Trading signal	Slide deck	Voice→PDF	Read text	OCR scan

Figure 4 — Read top-down: pick the job, read the row, use ✓ skills. This is the answer to 'future agents x pening nak cari' — one matrix, one look.

The 4 canonical jobs

A · Plain document (dossier/brief/report)

→ scientific-pdf-generation (Mode A/B) or forge-pdf-delivery. Backend: pandoc→xelatex (fast) or weasyprint (design).

B · Visual intelligence (briefing/dossier + charts)

→ scientific-pdf-generation Mode B/C + matplotlib. Backend: weasyprint HTML for the layout, matplotlib for figures embedded as base64.

C · Market chart → PDF

→ trading-signal-chart. Backend: matplotlib + reportlab table, single PdfPages.

D · Read/ingest an existing PDF

→ ocr-and-documents (text) / forge-document-intelligence (governed OCR + re-grounding) / pdf(productivity) (merge/split/form).

5 · Who owns what — agents, kernel, actuators

AAA agents — 15 registered, probed live

Agent ID	Role
opencode	forge-instrument
kimi-code	forge-instrument
claude-code	governed-agent
hermes-asi	autonomous-agent
aaa-gateway	operator
geox-witness	domain-specialist
wealth-witness	domain-specialist
geox-personal	analyst
arifos-kernel	governance
777-FORGE	witness-agent
antigravity	agentic-executor
codex-cli	forge-instrument
grok-build	forge-instrument
mesa-test-agent	test-agent
copilot-cli	governed-executor

arifOS kernel — 8 canonical tools, all live

The kernel is the gate. Every consequential artifact passes through: `arif_init` · `arif_observe` · `arif_think` · `arif_route` · `arif_memory` · `arif_judge` · `arif_forge` · `arif_seal`. **Authority ceiling: F13.** No agent self-authorizes.

A-FORGE actuators — 32 tools, the only hands

Actuators that touch a PDF artifact: `forge_execute` (runner) · `forge_filesystem` (read/write) · `forge_shell` (commands) · `forge_document_ingest` (PDF → structured) · `forge_chart` (chart render) · `forge_visual_qa` (verify before delivery).

Full surface: 32 tools. Fingerprint: 122 unique · 0 duplicates · 95 OBSERVE / 23 MUTATE (source: AForge_TOOL_AUDIT_2026-09-15.md).

6 · Live proof — the pipeline works end-to-end



Figure 5 — Captured from live WEALTH gold API (port 3456) at render time. XAUUSD 4183.41. The chart you are reading was drawn by matplotlib, embedded as PNG, composed with reportlab, and written to this PDF. That chain is the pipeline, not the description of it.

7 · Insert snippets — copy-paste-ready

A) HTML → PDF (weasyprint, design control)

```
from weasyprint import HTML
HTML('/tmp/poster.html').write_pdf('/tmp/poster.pdf')
```

B) Markdown → PDF (pandoc, fastest for text)

```
pandoc input.md -o out.pdf --pdf-engine=xelatex \
-V geometry:margin=11in --toc
```

C) Chart + text into ONE PDF (reportlab + matplotlib)

```
import matplotlib; matplotlib.use('Agg')
import matplotlib.pyplot as plt
from reportlab.platypus import (SimpleDocTemplate, Paragraph, Image)
fig, ax = plt.subplots(figsize=(11,4)); ax.plot([1,2,3],[1,4,9])
fig.savefig('/tmp/c.png', dpi=150)
doc = SimpleDocTemplate('out.pdf')
doc.build([Paragraph('Title', style), Image('/tmp/c.png', 17*cm, 6*cm)])
```

D) PDF → text (pymupdf, instant)

```
import pymupdf; doc = pymupdf.open('x.pdf')
for p in doc: print(p.get_text())
```

E) Visual QA before delivery

```
pdftoppm -png -r 110 out.pdf /tmp/pg
# then vision_analyze each /tmp/pg-N.png
```

F) Verify a real PDF (not mislabeled text)

```
file out.pdf # must say: PDF document, version 1.7
pdftinfo out.pdf | grep Pages
```

8 · Honest gaps — name them, don't fake them

Gap	Reality today	Movement
OCR scanned PDF (no text layer)	py-mupdf returns empty. marker-pdf not installed (~500MB, not on CI)	ObserveArtifact() if a URL exists; else install marker-pdf when di
Voice → PDF in one call	No engine does it. Requires chain: STT → markdown → weasyprint → PDF	Build weasyprint-Stack if requested; not urgent.
Live data re-render inside a built PDF	PDFs frozen at build time. No live-updating figure. Snapshot → freeze into HTML → render. Correct by design.	
Pixel-perfect modern CSS in PDF	WeasyPrint 70 partial on Grid. Chrome headless better. Chrome headless for modern layouts, weasyprint for speed	Use Chrome headless for modern layouts, weasyprint for speed
PDF embedding audio/video	reportlab supports attachments; no skill pattern documented	Added when needed; not a standing capability.

9 · The frame — why this matters

PDF is the visible handshake. Rule 11 of your own calibration: Telegram = meja hidangan; what lands is the actual meal. The PDF is the same object at a larger scale. An agent that produces a correct log nobody reads has not reached you. An agent that produces a broken PDF has reached you with something wrong. So the PDF skill is not a nice-to-have — it is the surface on which trust is either built or lost.

One flow, no skipping. The pipeline in Figure 1 is the same 9-node reality flow compressed into an artifact: signal observed, plan proposed, verified, judged, sealed, executed, witnessed as a finished PDF. When an agent skips from OBSERVE straight to EXECUTE (writes a PDF without verification), the artifact ships broken — and you get a 'Hang x bagi' reply. The gate is the gift.

Federate, don't proliferate. 13 skills scattered across two trees means the next agent guesses. The routing matrix (Figure 4) collapses that to one look. This dossier is not a new skill — it is a frame over the skills that already exist. Canon #0 three-test: it eliminates a demonstrated failure class (agent-scatter), compiles to a reference (the matrix), and improves a real decision (which skill for which job). It does not add machinery.